

Assignment 4

Step 1: Data Preparation

The dataset used for this analysis is the "City Lifestyle Segmentation Dataset," which contains data on 300 synthetic cities. The goal was to group these cities based on lifestyle factors such as income, rent, air quality, and happiness. Dataset: [link](#)

Data Cleaning: The dataset was found to be clean with **0 missing values**, so no rows needed to be dropped. However, the data contained columns with vastly different scales (e.g., `avg_income` ranges from 300 to 7,000, while `happiness_score` ranges from 1 to 10). To prevent the algorithm from being biased toward larger numbers, I applied **Feature Scaling (StandardScaler)** to normalize all numerical features before clustering.

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

# Load data and select only the numeric columns for math
df = pd.read_csv('city_lifestyle_dataset.csv')
X = df.select_dtypes(include=['number'])

# Scale the data (Crucial for K-Means to work correctly)
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

print("Step 1 Complete: Data loaded and scaled.")
```

Step 2: Visualization Before Clustering

Before applying the K-Means algorithm, I visualized the relationship between **Air Quality Index** and **Happiness Score** to observe any natural patterns.

Observation: The scatter plot revealed a negative correlation between the two variables: as air pollution increases (higher Air Quality Index), the happiness score tends to decrease. While the data points showed a clear trend, they appeared as a continuous cloud without clear, distinct groups visible to the naked eye. This confirms the need for a clustering algorithm to identify hidden segments in the multi-dimensional data.

```
# Plot Air Quality vs Happiness to see the raw data distribution
plt.figure(figsize=(10, 6))
plt.scatter(df['air_quality_index'], df['happiness_score'], color='gray',
            alpha=0.5)
plt.title('Step 2: Data Before Clustering')
plt.xlabel('Air Quality Index (Lower is Better)')
plt.ylabel('Happiness Score')
plt.grid(True)
plt.show()
```



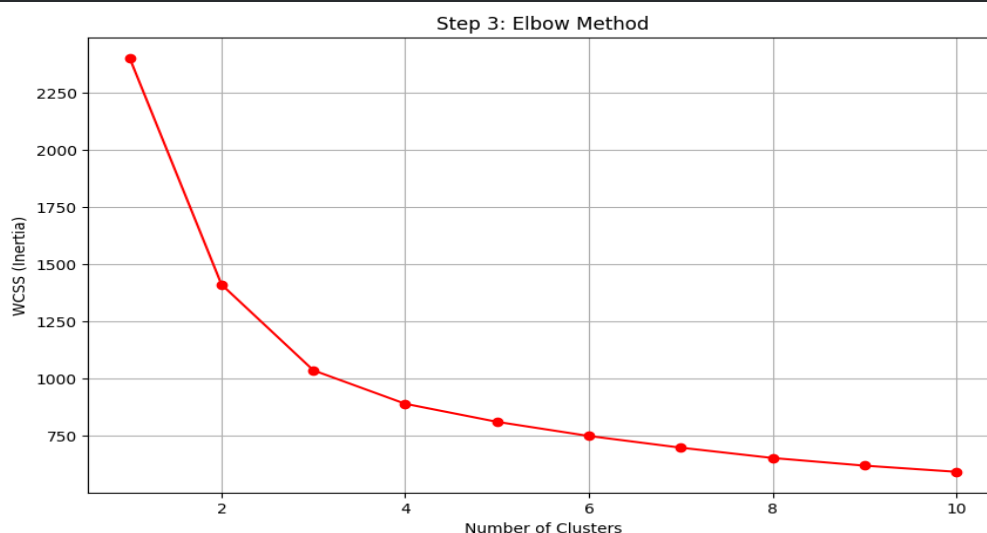
Step 3: Determining Optimal Clusters (Elbow Method)

To determine the optimal number of clusters (k), I utilized the **Elbow Method**. I ran the K-Means algorithm for k values ranging from 1 to 10 and calculated the Within-Cluster Sum of Squares (WCSS) for each.

Results: The plot of WCSS vs. Number of Clusters shows a sharp decline that begins to flatten out significantly at $k=5$. The "elbow" point indicates that adding more clusters beyond 5 yields diminishing returns in variance reduction. Therefore, $k=5$ was selected as the optimal number of clusters for this analysis.

```
# Test cluster counts from 1 to 10 to find the best 'k'
wcss = []
for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, init='k-means++', n_init=10,
random_state=42)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

# Plot the Elbow Curve
plt.figure(figsize=(10, 6))
plt.plot(range(1, 11), wcss, marker='o', color='red')
plt.title('Step 3: Elbow Method')
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS (Inertia)')
plt.grid(True)
plt.show()
```



Step 4: Visualization After Clustering

After assigning the cities to 5 clusters, I visualized the results. Since the dataset contains 8 dimensions (Income, Rent, Density, etc.), a simple 2D scatter plot often results in visual overlapping. To solve this, I used **Principal Component Analysis (PCA)** to project the 8-dimensional data into 2 principal components.

Observation: The PCA visualization demonstrates distinct, well-separated clusters. This confirms that the K-Means algorithm successfully partitioned the cities into meaningful groups based on their shared characteristics.

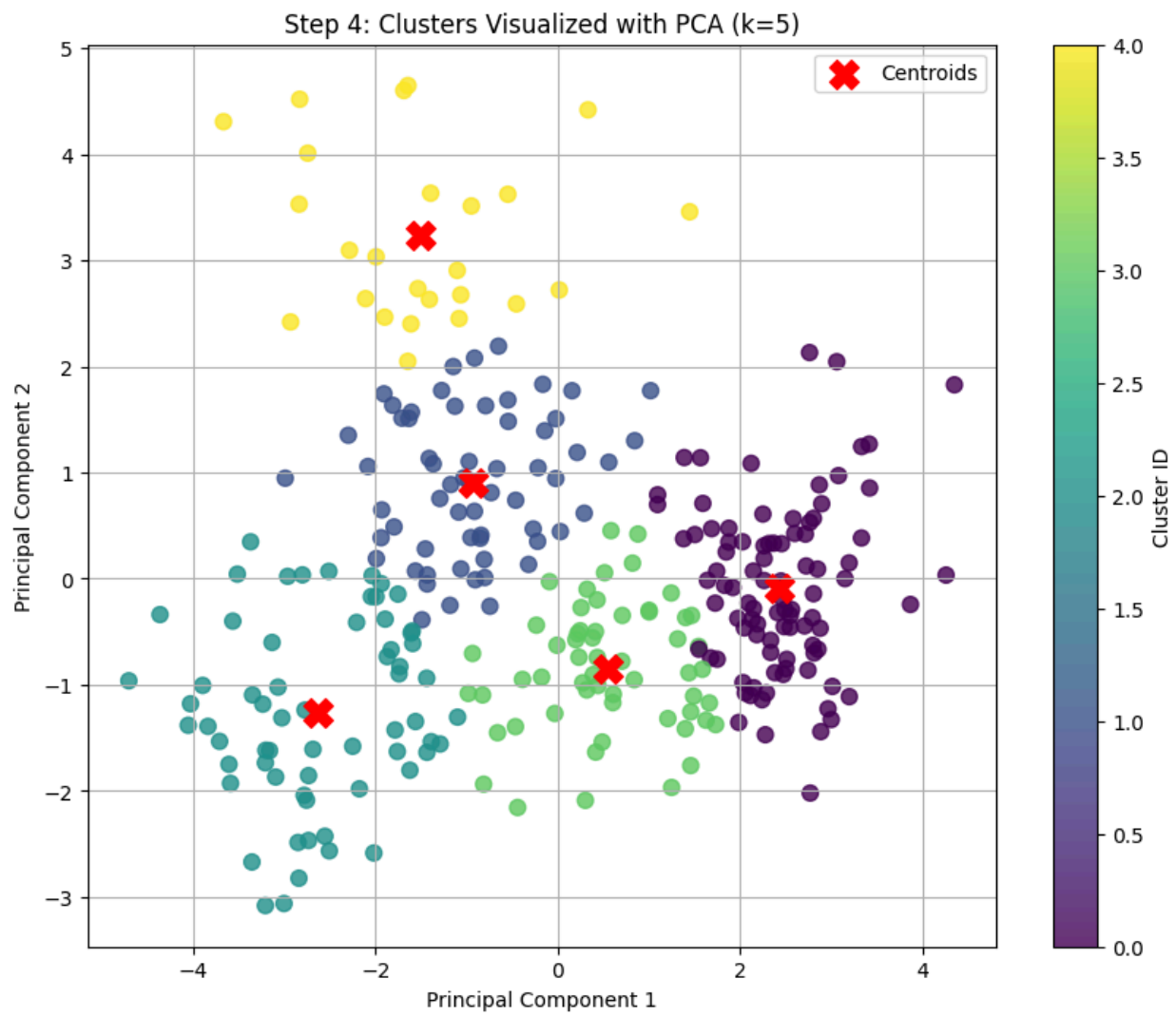
```
# 1. Run the final model with k=5 (based on the Elbow result)
optimal_k = 5
kmeans = KMeans(n_clusters=optimal_k, init='k-means++', n_init=10,
random_state=42)
df['cluster'] = kmeans.fit_predict(X_scaled)

# 2. Use PCA to project 8 dimensions down to 2 for a clean plot
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

# 3. Plot the distinct clusters
plt.figure(figsize=(10, 8))
scatter = plt.scatter(X_pca[:, 0], X_pca[:, 1], c=df['cluster'],
cmap='viridis', s=60, alpha=0.8)

# Add centroids (red X marks)
centroids_pca = pca.transform(kmeans.cluster_centers_)
plt.scatter(centroids_pca[:, 0], centroids_pca[:, 1], c='red', s=200,
marker='X', label='Centroids')

plt.title(f'Step 4: Clusters Visualized with PCA (k={optimal_k})')
plt.xlabel('Principal Component 1')
plt.ylabel('Principal Component 2')
plt.legend()
plt.colorbar(scatter, label='Cluster ID')
plt.grid(True)
plt.show()
```



Step 5: Interpretation of Results

Based on the average feature values for each group, the 5 identified clusters represent distinct city archetypes:

- **Cluster A: The Wealthy Tech Hubs**
 - **Characteristics:** Highest Average Income (~\$4,200), High Rent, and Excellent Internet Penetration (90%).
 - **Insight:** These are likely developed metropolitan areas with high costs of living but high quality of life (Happiness ~8.4).
- **Cluster B: The Industrial Mega-Cities**
 - **Characteristics:** Extreme Population Density (~10,000+), High Air Pollution (Index >115), and Low Happiness (~5.0).
 - **Insight:** These cities represent overcrowded industrial zones where environmental quality and life satisfaction are compromised.
- **Cluster C: The Developing Cities**
 - **Characteristics:** Lowest Income (~\$1,200), Lowest Internet Access (50%), and Lowest Happiness (~4.6).
 - **Insight:** These areas likely struggle with infrastructure and economic development.
- **Cluster D: The Eco-Friendly Towns**
 - **Characteristics:** Moderate Income, Good Air Quality, and High Happiness (~7.5) despite lower density.
 - **Insight:** These represent smaller, balanced cities that prioritize well-being over raw economic output.
- **Cluster E: The Standard Urban Centers**
 - **Characteristics:** Average scores across Density, Income, and Rent.
 - **Insight:** These are typical cities that do not lean towards any extreme but represent the "middle class" of urban development.